

# Adarsh Dwivedi

## DevOps and Platform Engineer

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Final-year CS engineer who treats reproducibility as the deliverable. Built Dockerized pipelines where the same inputs always produce the same outputs, a caching proxy on PyPI that fails open and reports itself degraded rather than failing requests, and services hardened with JWT RBAC that refuse to start without their encryption key. Deploys across AWS, Cloud Run, Vercel and Render, with CI that fails loudly rather than silently.

### EXPERIENCE

#### Deloitte India, Digital Excellence Centre

Jun 2026 – Aug 2026

AI Engineer (Agentic AI) Intern

Gurugram, India

- Built FastAPI and Pydantic backend services with schema-validated request and response contracts, powering tool-calling pipelines on an enterprise SAP-integrated platform.
- Root-caused a silent startup failure where an async Postgres driver deadlocked against Windows network I/O scheduling, evaluated every workaround, and selected the foundation the rest of the migration was built on.
- Authored the migration plan moving a production system onto new infrastructure without breaking existing frontend contracts, then validated both execution paths against a parity checklist before cutover.

### PROJECTS

#### PrimeTradeML, Reproducible MLOps Batch Pipeline | [GitHub](#)

Docker, Python, CI, YAML

- Built a deterministic, Dockerized batch pipeline for trading-signal generation where the same inputs always reproduce the same outputs, because a signal you cannot reproduce is a signal you cannot audit.
- Wired validation and observability into CI so a run fails loudly on a data or schema problem rather than emitting a quietly wrong signal, with per-stage timings recorded on every run.
- Kept configuration in YAML and the environment in the image, so a run is defined entirely by its inputs and its tag.

#### CacheLLM, Semantic Caching Proxy for LLM APIs (Published on PyPI) | [Live](#) | [GitHub](#)

Python, FastAPI, Docker, Redis, PyPI

- Shipped a caching proxy to PyPI fronting OpenAI, Gemini, Groq, Claude, Grok, Ollama and Bedrock behind one interface, routing each model to the host that serves it from that host's live model list.
- Made it run with nothing installed: the cache is an in-process numpy matrix by default with Redis optional, and the proxy fails open, forwarding upstream and reporting itself degraded rather than failing the request.
- Measured a 77% hit rate and 78% cost reduction, with a per-request policy deciding whether a response is safe to cache at all and for how long.

#### MedGuardX, Context-Aware PII and PHI Masking (Published on PyPI) | [Live](#) | [GitHub](#)

Python, FastAPI, Docker, JWT RBAC

- Shipped three layers, a stateless engine on PyPI, a hardened FastAPI service and a Next.js app, with the spaCy model size configurable so an operator picks the accuracy and memory trade-off at deploy time.
- Enforced authentication on every protected route with the role read from the JWT rather than the body, and refused privileged self-registration so admins are seeded out of band.
- Required the encryption key in production and made the service fail fast when it is missing, rather than silently falling back to a throwaway key.

### TECHNICAL SKILLS

**Infrastructure and DevOps:** Docker, AWS, Google Cloud Run, Vercel, Render, GitHub Actions, CI/CD, Linux, Git, Observability, Reproducible Builds

**Backend and APIs:** FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Next.js, React, Streamlit

**Languages:** Python, C/C++, Java, JavaScript, TypeScript, SQL

**Core Computer Science:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

### ACHIEVEMENTS

- Published three packages to PyPI, cachellm-proxy, creditsetu and medguardx-core, each installable, documented and covered by tests.
- 32 public repositories, most deployed and free to use, with every reported number reproducible from the repository and failures published beside results.

### EDUCATION

#### The LNM Institute of Information Technology, Jaipur

2023 – 2027

B.Tech, Computer Science and Engineering

CGPA: 7.11 / 10

### LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (Apr 2025 – Mar 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 – Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.